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Chapter 3: Research Methodology

3.1 Introduction

This chapter presents the research methodology used to investigate the detection of AI-generated steganographic malware embedded within digital images. The purpose of the methodology is to explain, in a clear and structured manner, how the study was conducted to achieve its research objectives covering the research approach, the design choices, how data was constructed and labelled, how detection was performed, and how results were evaluated (Sreekumar, 2023). Because steganographic malware is designed to appear benign and evade conventional detection (Mandal et al., 2022), the methodology must be rigorous and repeatable to ensure that the findings are defensible.

The study follows an approach where the primary aim is the development and evaluation of a hybrid detection artefact that combines signature-based detection with machine learning-based anomaly detection. The research is conducted using controlled datasets consisting of benign images and stego images generated through LSB-based embedding. The system does not run or open anything inside the image. Instead, it examines the image's raw bytes and extracted features to look for hidden or suspicious patterns before execution, making the analysis safe and non-invasive. The system's performance is evaluated using clear, quantifiable metrics, calculated by comparing its results against labels that are already known to be correct.

The remainder of this chapter is structured as follows. Section 3.2 talks about the research paradigm and approach, followed by Section 3.3 which outlines the research design. Section 3.4 defines the threat model and key assumptions that guide the scope of detection and evaluation. Section 3.5 explains the data sources and dataset used, and Section 3.6 details the steganographic sample generation process used to produce labelled stego images. The rest of the sections describe feature the YARA detection method, the anomaly detection method, and the hybrid fusion logic, before presenting the evaluation metrics, experimental procedure, tools and environment, and ethical and safety considerations. The chapter concludes with a summary that links the methodology to the system implementation presented in Chapter 4.

3.2 Research Paradigm and Approach

Research paradigms represent the fundamental beliefs that guide how a researcher understands reality, knowledge creation, and the process of inquiry (Maruashvili, 2024). These beliefs are shaped by ontological, epistemological, methodological, and axiological assumptions, which collectively influence how research problems are framed and investigated (Vaishnavi and Kuechler, 2004). Understanding the research paradigm is essential, as it determines how reality is perceived and how knowledge about that reality can be systematically studied (Rehman and Alharthi, 2016).

This study follows a quantitative and experimental research approach, where controlled datasets of benign and steganographic images are evaluated using measurable detection performance metrics, and furthermore, the choice of paradigm directly affects how threats are modelled, how detection mechanisms are designed, and how results are validated (Kotronoulas and Papadopoulou, 2023). This study considers three paradigmatic perspectives - positivism, interpretivism, and design science research to determine the most suitable method for investigating the detection of AI-generated steganographic malware.

3.2.1 Positivism

Positivism assumes the existence of an objective reality that can be observed, measured, and analysed through empirical and quantitative methods. Within this paradigm, research aims to uncover causal relationships and generalisable laws using controlled experimentation and statistical evaluation (Bryman, 2016).

This paradigm aligns partially with the objectives of this study, as the detection of steganographic malware can be measured objectively using metrics like accuracy, precision, recall, and error rates. The study adopts a positivist stance when measuring the effectiveness of detection techniques, treating malware detection outcomes as observable and repeatable phenomena independent of human interpretation. By testing the system on known clean images and images with hidden data, the study shows how LSB embedding affects detection performance.

However, positivism alone is insufficient to fully address the research problem, as it does not inherently support the design and construction of new detection artefacts. While positivist principles guide the evaluation phase of this research, they do not fully capture the creative and problem-solving aspects required to design a novel hybrid detection system (Park et al., 2020).

3.2.2 Interpretivism

Interpretivism views reality as socially constructed and shaped by human experiences, perceptions, and contextual understanding. Rather than seeking universal laws, interpretivist research aims to understand meaning, intent, and context, often relying on qualitative methods such as interviews, observations, and textual analysis (William, 2024).

In this study on steganographic malware detection, interpretivism has limited applicability. The detection of hidden malicious content in digital images does not rely on interpretation but rather on statistical, structural, and algorithmic properties of the files. The presence or absence of steganographic malware is not dependent on human perception but on measurable deviations within the digital artefacts.

As a result, interpretivism is not adopted as the primary paradigm for this study. Nevertheless, interpretivist insights indirectly inform the understanding of how attackers design evasive malware techniques and how defenders conceptualise detection challenges. These contextual insights do support the motivation for the research but do not form the methodological foundation.

3.2.3 Design Science Research

Design Science Research focuses on solving real-world problems through the creation, evaluation, and refinement of artefacts such as models, methods, frameworks, or systems (Hevner et al., 2004). The primary objective of DSR is to bridge the gap between theory and practice by producing artefacts that are both theoretically grounded and practically applicable.

This study is fundamentally focused on the Design Science Research paradigm as its primary research paradigm, supported by a positivist approach for empirical evaluation. The core contribution of the research is the design and evaluation of a hybrid detection artefact that integrates YARA-based signature detection with machine learning-based anomaly detection to identify AI-generated steganographic malware in digital images. The artefact is not conceptual, but it is implemented as a working system that can be tested and measured using actual experiments.

DSR is particularly appropriate for this study because existing antivirus solutions and signature-based tools struggle to detect malware concealed using modern steganographic techniques (Hussain Hussain, 2025). By building a new detection pipeline, the study tackles a specific cybersecurity problem and produces practical, usable insights for both research and real-world use. The approach also allows the system to be tested, improved, and compared against existing methods over time.

3.3 Research Design

Research design defines the structured plan and procedures used to address the research objectives and answer the research questions of a study (Creswell and Creswell, 2017). In this study, the design is formulated to support the systematic development, implementation, and evaluation of a detection system for AI-generated steganographic malware embedded within digital images.

The study follows a Design Science Research approach, focusing on building a working detection system and evaluating how well it performs. The research design is driven by the need to address a clearly identified cybersecurity problem: the inability of traditional antivirus and signature-based detection systems to reliably detect malware concealed using modern steganographic techniques as described by Alazab and Pham (2023). Rather than observing existing systems or user behaviour, the study focuses on building and empirically validating a hybrid detection pipeline.

The research design further integrates artefact construction with quantitative experimental evaluation, ensuring that the proposed detection system is not only theoretically grounded but also practically assessable. The artefact is evaluated using controlled datasets of benign and steganographic images, allowing detection performance to be measured using the standard metrics such as accuracy, precision, recall, false positive rate, and false negative rate.

The Design Science approach is selected due to its suitability for addressing complex, real-world problems through the creation of innovative and testable solutions (Hevner et al., 2004). The study results in a working detection system that combines YARA rules with machine-learning anomaly detection, allowing the system to be tested systematically and compared with existing baseline methods.

The study is guided by Design Science Research principles, and its evaluation relies on objective, measurable evidence.

Belief	Positivism	Interpretivism	Design Science
Ontology	A single objective reality independent of observation	Multiple realities constructed through social interaction	Multiple but shared realities shaped by problem context and artefact use

Epistemology	Knowledge discovered through objective measurement	Knowledge derived from subjective interpretation	Knowledge created through building and evaluating artefacts
Methodology	Controlled experiments and measurement	Qualitative interpretation and analysis	Artefact development, experimentation, and evaluation
Axiology	Value-neutral inquiry	Contextual and value-dependent	Utility, effectiveness, and problem-solving value

Table 1: Summarises how the adopted research design aligns with common research paradigms.

As noted within this design, the system is built and improved using Design Science, while its performance is tested objectively and in a repeatable way using measurable evidence. Interpretivism is excluded because the study does not depend on human opinions or subjective analysis.

3.3.1 Experimental Design and Detection Perspective

The experimental design of this study is structured to reflect realistic antivirus scanning scenarios while maintaining strict control over variables to ensure reliable evaluation. The study adopts a black-box detection perspective, where images are analysed solely based on their observable byte-level and statistical properties. No assumptions are made about the internal logic or execution behaviour of embedded payloads. This perspective aligns with real-world antivirus scanning, where files are often analysed without execution.

3.3.2 Static Analysis Focus

The system only analyses images offline using static techniques. Nothing inside the images is run or executed, which matches how many antivirus tools work and avoids the risk of triggering malware. The goal is to detect hidden steganographic signs before anything can run.

3.3.3 Image-Based Steganographic Scope

The study focuses exclusively on **image-based steganographic malware**, specifically using Least Significant Bit embedding techniques. Images are selected as the carrier medium due to their high prevalence, inherent redundancy, and demonstrated effectiveness as steganographic containers in prior research (Morkel et al., 2005). Other

carrier formats such as audio, video, or document files are excluded to maintain experimental control and depth of analysis.

3.3.4 Scope and Boundaries

The scope of the research is limited to:

- Static detection of steganographic malware in images
- LSB-based embedding techniques
- Signature-based and anomaly-based detection

The study explicitly excludes:

- Runtime malware execution
- Behaviour-based or sandbox analysis
- Polymorphic or self-modifying payload execution
- Non-image steganographic carriers

These limits help keep the evaluation clear, consistent, and scientifically reliable, so the detection system can be tested properly and repeated with the same results.

3.4 Threat Model and Assumptions

The study uses a realistic threat model based on how steganographic malware is described in current research. It clearly defines what attackers can do, what defenders can detect, the limits of antivirus tools, and what attack scenarios are not covered. This keeps the evaluation controlled, realistic, and relevant to real-world antivirus settings.

3.4.1 Attacker and Defender Capabilities

From the attacker's perspective, it is assumed that adversaries have the capability to embed malicious payloads within digital images using steganographic techniques. Prior studies have shown that attackers can successfully conceal malware within benign multimedia files, enabling such threats to evade traditional antivirus detection mechanisms as shown by Mirza et al. (2021) and Almomani et al. (2022). The attacker is assumed to select or generate benign image files as carrier media in this study and embed payloads using spatial-domain steganography, particularly Least Significant Bit substitution.

From the defender's side, the study assumes a static analysis setup similar to many antivirus tools, where image files are examined before execution using inspection methods, without running or unpacking any hidden payloads. Detection capabilities are therefore limited to byte-level analysis, statistical inspection, and rule-based matching mechanisms such as YARA (Greco and Ianni, 2025). This defensive model aligns with common antivirus practices, where files are scanned offline to prevent malicious execution before damage occurs (Kumar et al., 2023).

3.4.2 Assumptions About Steganographic Techniques and Antivirus Limitations

The study assumes malware is hidden in images using spatial-domain methods, mainly Least Significant Bit substitution. LSB is chosen because it is well known in research for hiding data without visibly changing the image, making the concealment hard to notice (ALabaichi et al., 2020). It is assumed that payloads are distributed across image pixels, do not significantly alter file size or format, and payloads are unlikely to appear as contiguous byte sequences.

The threat model also assumes that traditional antivirus systems have built-in weaknesses when trying to detect malware hidden inside multimedia files like images. Studies have demonstrated that while antivirus engines may detect malware in its original form, detection rates decrease significantly once the same payload is concealed within an image or video frame (Mirza et al., 2021). Using signatures alone is not enough, because steganography can break up or hide known patterns. This is why the study also uses anomaly detection, which looks for unusual statistical patterns caused by steganographic changes (Hammadeh and Kavitha, 2023).

3.4.3 Excluded Threat Scenarios

To maintain a focused and controlled experimental scope, several threat scenarios are explicitly excluded from this study. The proposed detection system does not address runtime extraction or execution of embedded payloads, behaviour-based or sandbox-based malware analysis, polymorphic or self-modifying malware loaders, or encrypted multi-stage droppers. By excluding these threats, the study ensures that the evaluation remains aligned with its objective of assessing static detection capabilities for AI-generated steganographic malware in digital images.

3.5 Data Sources and Dataset Construction

This section describes the data sources and dataset construction strategy used in the study to evaluate the detection of AI-generated steganographic malware in digital images. A carefully controlled dataset was needed so experiments could be repeated, labels would

be reliable, and performance could be measured objectively. Instead of using external datasets, the study created its own dataset to fully control how data was embedded, what payloads looked like, and the testing conditions.

3.5.1 Dataset Categories

The dataset used in this study consists of two primary categories of images:

Benign (Original) Images

Original images represent clean, non-malicious input files. These images serve as the baseline against which steganographic modifications are evaluated. Original images are uploaded into the system and stored without any embedded payloads. They are labelled as benign and used for training and evaluation of detection techniques, particularly for establishing normal statistical patterns during anomaly detection.

Stego (Malicious) Images

Stego images are generated from the original images by embedding payload data using Least Significant Bit substitution. These images simulate steganographic malware carriers and are labelled as simulated malicious carriers. Each stego image is explicitly linked to its corresponding original image, ensuring traceability and enabling controlled comparison between benign and modified files.

3.5.2 Image Formats and Constraints

All images used in the study are processed in PNG format. PNG was selected to avoid lossy compression, which could destroy or distort embedded payload data and interfere with detection accuracy (Roelofs, 2003). Images are converted to RGB format prior to embedding to ensure consistent handling of colour channels and embedding capacity. Image size and resolution were limited so there was enough space to hide data while keeping stego images visually similar to the originals. This also helps keep feature extraction consistent and avoids issues caused by very small or very large images.

3.5.3 Payload Types and Sizes

Payloads embedded within stego images are AI-generated, non-executable data sequences to ensure ethical, safe, and controlled experimentation.

The study utilises:

- **AI-generated text and byte-like payloads**, produced to simulate diverse and previously unseen content patterns for controlled steganographic detection evaluation

- The EICAR antivirus test string and additional AI-generated synthetic payload sequences, used to model variability and unpredictability commonly associated with adaptive or evolving steganographic malware.

Payload sizes are limited to what each image can safely hold. Checks are done before embedding to ensure the available least significant bits are not exceeded, which prevents image corruption and keeps experiments consistent.

3.5.4 Ground Truth Labelling Strategy

Labels are assigned upfront when the dataset is created. Clean images are marked benign, and images with embedded payloads are marked stego. The system does not guess these labels during detection. This makes performance evaluation objective and reliable, allowing metrics like accuracy, precision, recall, and error rates to be calculated correctly.

3.5.5 Dataset Separation

To support valid experimental evaluation, the dataset is logically separated into distinct subsets:

- **Training set:** Used to train the machine learning–based anomaly detection model, typically consisting of benign images to establish a baseline of normal behaviour
- **Testing set:** Used to evaluate detection performance on unseen images
- **Benchmarking set:** Used to compare the performance of individual detection stages (YARA-only, anomaly detection-only) against the hybrid detection pipeline

These separations are enforced within the experimental workflow to prevent data leakage and ensure unbiased evaluation.

3.6 Steganographic Sample Generation Methodology

Steganographic samples in this study are generated using Least Significant Bit (LSB) substitution, selected due to its prevalence in the literature and its effectiveness in concealing payloads while preserving visual quality as explained in the previous chapter (Literature Review). LSB substitution alters the least significant bits of pixel values, introducing minimal perceptual distortion, which makes it suitable for simulating realistic steganographic malware scenarios and challenging static detection mechanisms.

Payloads embedded within images are non-executable as noted, including AI-generated and controlled synthetic byte strings used to simulate unknown payload patterns. Payload generation is deterministic and controlled to ensure repeatability across experiments.

The study does not consider other hiding methods like EXIF metadata changes, appending data to files, or abusing PNG file chunks. These methods usually create file-structure anomalies instead of changing pixel data. Since the detection system focuses on pixel-level steganography, those techniques are intentionally left out (Koptyra and Ogiela, 2023). The chosen LSB approach aligns with the passive warden model, where the defender observes files without modifying them and detection is performed solely through static analysis (Stacey, 2014).

3.7 Feature Engineering and Analysis Strategy

Feature engineering is used to detect hidden statistical changes caused by steganography that cannot be seen visually. Because the system uses static analysis, features are taken directly from the image files to capture changes in data patterns created by embedded payloads.

The study uses features based on entropy, byte frequency, LSB irregularities, and image structure. Entropy measures increased randomness, byte statistics detect unusual value distributions, LSB features look for abnormal bit patterns, and structural features capture file-level properties relevant to static analysis.

All features are normalised so they can be compared fairly, and no single feature dominates the detection. The same feature set is applied to all images, ensuring consistent representation and reliable anomaly detection.

3.8 Signature-Based Detection Methodology (YARA)

YARA is employed as the signature-based detection in the detection pipeline, serving as the first stage of analysis. Its role is to identify known indicators of malicious, or test payloads embedded within image files through pattern matching which will be created in the study.

YARA rules are designed following a specific rule design, targeting specific byte-level indicators rather than overly general patterns. The rules focus on detecting known string patterns where identifiable indicators remain present; however, purely LSB-based embedding is expected to significantly reduce direct signature visibility.

While YARA provides fast and interpretable detection for known indicators, it has inherent limitations when applied to LSB steganographic images. As a result, YARA is expected to exhibit limited detection capability for purely LSB-based stego images, reinforcing the need for complementary anomaly-based detection.

3.9 Machine Learning–Based Anomaly Detection Methodology

Anomaly detection is employed to identify steganographic images that evade signature-based detection. An unsupervised learning approach is selected due to the limited availability of labelled steganographic malware samples and the evolving nature of embedding techniques.

The study uses **Isolation Forest** as the anomaly detection algorithm due to its effectiveness in identifying outliers within high-dimensional feature spaces (Xu et al., 2023). The model operates by isolating anomalies based on how easily samples can be separated from the majority of data.

The training strategy follows a **per-scan, dynamic training approach**, where the model is trained using the available feature data prior to inference. Anomaly scores produced by the model represent the degree of deviation from normal image characteristics. Thresholds are selected empirically to distinguish between benign and anomalous samples.

Compared to supervised classifiers, this approach does not require extensive labelled datasets and is better suited to identifying statistical deviations consistent with previously unseen steganographic patterns, making it appropriate for the study's objectives.

3.10 Hybrid Detection Pipeline Design

The detection system follows a two-stage hybrid pipeline consisting of signature-based detection followed by anomaly-based detection. Images are first scanned using YARA, and only those not flagged as malicious proceed to the anomaly detection stage.

Decision fusion is performed sequentially, where a positive YARA match immediately classifies an image as malicious. If no signature match is found, the anomaly detection result is used to determine whether the image exhibits steganographic characteristics. Conflicting outcomes are resolved by prioritising detection confidence while minimising false positives.

This sequential design reduces unnecessary computation and limits false positive propagation. Parallel execution is avoided to preserve interpretability and maintain a clear detection decision path consistent with static antivirus workflows.

3.11 Evaluation Metrics and Performance Criteria

Detection performance is evaluated using standard quantitative metrics with **accuracy** measuring the overall classification correctness, while precision and recall assess detection reliability and sensitivity, respectively. False positive rate and false negative rate

are used to evaluate misclassification behaviour, which is particularly important in security-sensitive contexts.

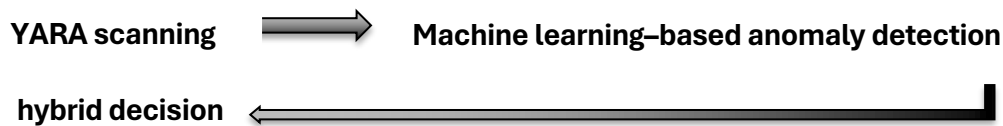
Comparative benchmarking is conducted by evaluating the performance of YARA-only detection, anomaly detection-only, and the hybrid detection pipeline. This comparison allows the contribution of each detection stage to be assessed independently.

Accuracy alone is insufficient as a performance measure, as it may mask poor detection of minority classes. Therefore, multiple metrics are used to provide a balanced and meaningful evaluation of detection effectiveness.

3.12 Experimental Procedure

The experimental procedure follows a fixed and repeatable workflow. A scan workspace is initialised, after which images are uploaded into the system. For stego samples, payload embedding is performed prior to detection.

Detection is executed in the following order:



Results from each stage are recorded, including detection outcomes and anomaly scores. All results and associated metadata are stored persistently to support evaluation, comparison, and reproducibility. This structured procedure ensures consistency across experimental runs and enables reliable performance analysis.

3.13 Tools, Technologies, and Environment

This section describes the tools, technologies, and execution environment used to design, implement, and evaluate the proposed steganographic malware detection system. The selected tools support static analysis, controlled experimentation, and reproducible evaluation of detection performance. All technologies employed in this study were chosen based on their suitability for image-based malware analysis, integration capability, and alignment with real-world antivirus scanning scenarios.

3.13.1 Software Tools

The detection system was implemented using **Python** as the primary programming language. Python was selected due to its strong ecosystem for malware analysis, machine learning, and image processing, as well as its compatibility with YARA and cloud deployment environments.

The backend detection service was implemented using **FastAPI**, a modern Python web framework. FastAPI was used to expose RESTful endpoints for image upload, steganographic embedding, scanning, and result retrieval. Its asynchronous execution model and lightweight design make it suitable for handling concurrent scan requests during experimentation.

For development and testing, the system was executed in a standard desktop environment Windows 11 Pro prior to cloud deployment. Version control was managed using **Git**, enabling controlled iteration and traceability of changes to the detection system throughout the research process.

3.13.2 Libraries and Frameworks

Several specialised Python libraries were used to support the detection pipeline:

- **YARA (yara-python)** was used to implement signature-based detection. YARA rules were compiled and executed directly on image byte streams to identify known steganographic indicators and embedded payload signatures.
- **scikit-learn** was used to implement the machine learning-based anomaly detection component. The Isolation Forest algorithm was employed to identify statistical anomalies in image features without requiring labelled malicious training data.
- **NumPy** was used for numerical operations and feature vector manipulation.
- **Pillow (PIL)** was used for image loading, conversion, and pixel-level access required for steganographic embedding and feature extraction.
- **joblib** was used for managing model-related operations during anomaly detection workflows.

These libraries collectively enabled static analysis of images, feature extraction, and detection logic without executing any embedded payloads.

3.13.3 Cloud Infrastructure

The backend detection system was containerised using **Docker** and deployed on **Google Cloud Run**, a serverless container execution platform. Cloud Run was selected to provide a consistent and controlled runtime environment that mirrors real-world deployment scenarios for malware scanning services.

Cloud Run allows the detection system to scale automatically based on request volume while maintaining isolation between scans. Environment variables were used to manage configuration values securely, ensuring that sensitive information such as API keys was not hardcoded into the application.

The frontend interface was hosted using **Firebase Hosting**, enabling users to upload images, configure scans, and view detection results through a web-based interface. Communication between the frontend and backend occurs via secure HTTPS endpoints.

3.13.4 Storage and Database Role

The system uses **Firebase Storage** and **Cloud Firestore** as integral components of the experimental workflow.

- **Firebase Storage** is used to store all uploaded images, including both benign (original) images and steganographic (malicious) images. Images are retrieved from storage into memory during scanning, ensuring that detection operates on raw byte streams without reliance on local disk storage.
- **Cloud Firestore** is used to persist structured metadata and detection results. This includes image identifiers, ground-truth labels, YARA detection outcomes, anomaly scores, hybrid detection decisions, and evaluation metrics.

This storage design supports dataset traceability, separation of experimental runs, and repeatable analysis. By persisting scan results, the system allows subsequent evaluation and benchmarking without requiring repeated execution of earlier detection stages.

3.13.5 Reproducibility Considerations

Reproducibility is a core requirement of this study and is supported through several design decisions:

- All steganographic samples are generated within the system using controlled embedding parameters.
- Original and stego images are explicitly linked through metadata stored in Firestore.
- Detection stages (YARA, anomaly detection, hybrid decision) are executed in a fixed and documented order.
- Feature extraction and anomaly detection operate deterministically given the same inputs.
- Cloud deployment ensures consistent runtime behaviour across experimental executions.

These measures ensure that experimental results can be reproduced and validated under the same conditions, supporting both academic rigor and future extension of the research. The tools, technologies, and environment used in this study were carefully selected to

support the design, implementation, and evaluation of a hybrid steganographic malware detection system. By leveraging Python-based analysis, YARA signature scanning, machine learning-based anomaly detection, and cloud-hosted infrastructure, the study ensures controlled experimentation, repeatability, and practical relevance to modern antivirus detection environments.

3.14 Validity, Reliability, and Limitations

The internal validity of this study is supported through the use of controlled datasets, deterministic steganographic embedding, and a fixed execution order for all detection stages. Ground-truth labels are assigned at dataset creation, ensuring that detection outcomes are directly attributable to the proposed hybrid detection system rather than external factors. Construct validity is maintained by evaluating detection performance using standard and widely accepted metrics, including accuracy, precision, recall, and error rates, which directly measure the system's ability to identify steganographic manipulation in images.

External validity is limited by the controlled experimental scope of the study. The dataset is synthetically generated and focuses on image-based steganography using Least Significant Bit embedding, which restricts direct generalisation to other steganographic techniques or carrier formats. In addition, the detection system operates exclusively through static analysis and does not address runtime extraction or behavioural malware analysis, which further constrains applicability to pre-execution detection scenarios.

Despite these limitations, the results are generalisable within the defined scope of static detection of image-based steganographic malware. The methodology and hybrid detection framework provide a reproducible and extensible foundation that can be adapted in future work to include additional embedding techniques, datasets, and detection strategies.

3.15 Ethical and Safety Considerations

This study involves the analysis of steganographic techniques and malware detection mechanisms, which raises important ethical and safety considerations. To ensure that the research does not introduce security risks, violate ethical standards, or cause unintended harm, a number of precautionary measures were adopted throughout the design, implementation, and evaluation phases of the study.

3.15.1 Malware Handling Precautions

To ensure the safety of the research environment and to prevent the accidental release or execution of harmful malware, no real-world malicious binaries were used in this study. Instead, the research exclusively utilised the AI-generated non-executable synthetic

payload strings, including AI-generated synthetic test strings, as well as the EICAR antivirus test string, which is widely used as a safe malware simulation standard.

The EICAR string is specifically designed for antivirus testing and does not perform any malicious actions. Its use allows the simulation of malware detection behaviour without introducing executable threats or dangerous processes into the research environment. In this study, the EICAR string and similar benign AI-generated string-based payloads were embedded into images to evaluate detection capabilities.

AI-generated payloads used in the experiments were limited to non-executable string-based data, formatted in a manner similar to the EICAR test string. These payloads were not designed to function as real malware and do not violate ethical or legal boundaries related to malware creation or distribution.

3.15.2 No Execution of Malicious Payloads

At no point during the study were embedded payloads executed, unpacked, or interpreted as runnable code. The detection system operates entirely through **static analysis**, examining image files at the byte and statistical level. This design choice eliminates the risks associated with dynamic malware execution, such as unintended system compromise or network propagation.

By avoiding payload execution, the study ensures that all experiments remain within a controlled and safe analytical context, consistent with ethical cybersecurity research practices.

3.15.3 Controlled Dataset Usage

All datasets used in this study were generated and managed within a **controlled experimental environment**. Benign images and their corresponding steganographic variants were explicitly labelled and linked through metadata, ensuring traceability and preventing accidental misuse.

The datasets were used solely for academic research purposes and were not shared publicly in a form that could facilitate misuse. Access to stored images and scan results was restricted to the research environment, and experimental runs were isolated to prevent cross-contamination between datasets.

3.15.4 Data Privacy Considerations

The study did not involve human participants, personal data, or sensitive user information. All images used for experimentation were non-personal and did not contain identifiable information. Metadata stored in the database was limited to technical attributes required

for detection and evaluation, such as file identifiers, detection results, and performance metrics.

By excluding personal or sensitive data, the study complies with data protection principles and avoids ethical concerns related to privacy or consent.

3.15.5 Responsible Disclosure Boundaries

The purpose of this research is to improve detection mechanisms for steganographic malware, not to enable the development or deployment of malicious techniques. Detailed implementation of embedding and detection methods is presented strictly within an academic context and does not include instructions for real-world exploitation.

The study does not disclose vulnerabilities in specific commercial antivirus products, nor does it provide actionable guidance for bypassing security systems. Findings are reported at a conceptual and evaluative level to support defensive cybersecurity research and responsible knowledge advancement.

3.16 Chapter Summary

This chapter presented the methodological choices adopted in this study to investigate the detection of AI-generated steganographic malware in digital images. The research was conducted within a Design Science Research paradigm and followed a quantitative and experimental approach. The chapter justified the research design, defined the scope and boundaries of the study, and described the experimental setup, dataset construction, steganographic sample generation, feature engineering strategy, and detection methodologies. The selected tools, technologies, and cloud environment were outlined to support controlled experimentation, reproducibility, and objective evaluation, while ethical and safety considerations ensured responsible handling of simulated malware.

The methodology established in this chapter provides the foundation for the system described in the following chapter. Chapter 4 presents the system implementation, detailing how the proposed hybrid detection approach was realised in a working system, including the architecture, components, and operational workflow used to translate the methodological design into practice.

References

- ALABAICHI, A., AL-DABBAS, M. A. A. K. & SALIH, A. 2020. Image steganography using least significant bit and secret map techniques. *International journal of electrical & computer engineering (2088-8708)*, 10.
- ALAZAB, M. & PHAM, T. 2023. Stegomalware: A systematic survey of malware hiding and detection in images, machine learning models and research challenges. *Authorea Preprints*.

- ALMOMANI, I., ALKHAYER, A. & EL-SHAFI, W. 2022. Novel Ransomware Hiding Model Using HEVC Steganography Approach. *Computers, Materials & Continua*, 70.
- BRYMAN, A. 2016. *Social research methods*, Oxford university press.
- CRESWELL, J. W. & CRESWELL, J. D. 2017. *Research design: Qualitative, quantitative, and mixed methods approaches*, Sage publications.
- GRECO, C. & IANNI, M. 2025. A cross-architecture malware detection approach based on intermediate representation. *Journal of Information Security and Applications*, 93, 104117.
- HAMMADEH, K. & KAVITHA, M. 2023. Unraveling ransomware: Detecting threats with advanced machine learning algorithms. *International Journal of Advanced Computer Science and Applications*, 14.
- HEVNER, A., R, A., MARCH, S., T, S., PARK, PARK, J., RAM & SUDHA 2004. Design Science in Information Systems Research. *Management Information Systems Quarterly*, 28, 75.
- HUSSAIN HUSSAIN, R. A. 2025. Evaluating antivirus software: a comparative analysis of detection methodologies and performance metrics in modern antivirus solutions.
- KOPTYRA, K. & OGIELA, M. R. 2023. An Efficient Steganographic Protocol for WebP Files. *Applied Sciences*, 13, 12404.
- KOTRONOULAS, G. & PAPADOPOULOU, C. A primer to experimental and nonexperimental quantitative research: the example case of tobacco-related mouth cancer. *Seminars in Oncology Nursing*, 2023. Elsevier, 151396.
- KUMAR, J., RAJENDRAN, B. & SUDARSAN, S. D. 2023. Zero-day malware classification and detection using machine learning. *SN Computer Science*, 5, 93.
- MANDAL, P. C., MUKHERJEE, I., PAUL, G. & CHATTERJI, B. 2022. Digital image steganography: A literature survey. *Information sciences*, 609, 1451–1488.
- MARUASHVILI, E. 2024. The Most Common Research Paradigms. *Georgian Academy of Business Sciences" Moambe"*, 14–17.
- MIRZA, S., ABBAS, H., SHAHID, W. B., SHAFQAT, N., FUGINI, M., IQBAL, Z. & MUHAMMAD, Z. A malware evasion technique for auditing android anti-malware solutions. 2021 IEEE 30th International Conference on Enabling Technologies: Infrastructure for Collaborative Enterprises (WETICE), 2021. IEEE, 125–130.
- MORKEL, T., ELOFF, J. H. & OLIVIER, M. S. An overview of image steganography. Issa, 2005. 1–11.
- PARK, Y. S., KONGE, L. & ARTINO JR, A. R. 2020. The positivism paradigm of research. *Academic medicine*, 95, 690–694.
- REHMAN, A. A. & ALHARTHI, K. 2016. An introduction to research paradigms. *International journal of educational investigations*, 3, 51–59.
- ROELOFS, G. 2003. PNG Lossless Image Compression. *Lossless Compression Handbook*, 371–390.
- SREEKUMAR, D. 2023. What is research methodology? Definition, types, and examples. *Paperpal Blog-Academic Writing Guides*.
- STACEY, D. Passive warden using statistical steganalysis. Proceedings of the 3rd annual conference on Research in information technology, 2014. 27–32.
- VAISHNAVI, V. & KUECHLER, W. 2004. Design research in information systems.
- WILLIAM, F. K. A. 2024. Interpretivism or constructivism: Navigating research paradigms in social science research. *Interpretivism or Constructivism: Navigating Research Paradigms in Social Science Research*, 143, 5–5.
- XU, H., PANG, G., WANG, Y. & WANG, Y. 2023. Deep isolation forest for anomaly detection. *IEEE Transactions on Knowledge and Data Engineering*, 35, 12591–12604.